

Elbow and arm pain

with BILL VICENZINO, ALEX SCOTT, SIMON BELL and NEBOJSA POPOVIC

Throwing warmup pitches to start the bottom of the second inning, Hudson felt some tightness around his elbow. On the second pitch that inning, he felt that something was seriously wrong. His fastball dipped to about 85 mph, and he became hittable. He stayed in the game and completed the inning . . . but that . . . would be Hudson's last pitch for another year.

Report on Major League Baseball pitcher Daniel Hudson's return from two ulnar collateral ligament surgeries

A well-functioning elbow is essential for upper limb use in sports. Elbow injuries may even interfere with an athlete's everyday activities. Tennis was the classic cause of elbow pain, but the double-handed backhand has reduced the prevalence of 'tennis elbow' dramatically. Elbow pain remains a problem in golf and in sports such as volleyball and handball, which involve forceful elbow hyperextension.

To approach elbow pain in clinical practice, consider which of the following categories describes your patient's elbow pain:

- lateral elbow pain, with a particular focus on extensor tendinopathy
- medial elbow pain
- posterior elbow pain
- acute elbow injuries
- forearm pain
- upper arm pain.

ANATOMY

The elbow is situated between two highly mobile joints, the shoulder and the wrist, and comprises three distinct articulations—the ulna-humeral, radio-capitellar and proximal radio-ulnar joints. The first two provide flexion (150°) and extension (0°) of the elbow and the third provides pronation/supination (85°/85°) of the forearm. The proximal radio-ulnar joint works in conjunction with the distal

radio-ulnar joint at the wrist to achieve forearm pronation-supination. With the forearm in full extension and supination there will be a physiological valgus (carrying angle) of 9–14° in men or 12–17° in women. Any increase or loss of this physiological angle is a sign of pathology (instability, mal-union or overuse). Elbow stability is provided by the osseous anatomy, capsuloligamentous structure and musculotendinous units that cross the elbow.

Ligaments

The ligaments of the elbow are divided into lateral and medial collateral ligament complexes. The medial collateral ligament complex is composed of discrete bands; the lateral collateral ligament complex consists of complex ligament fibres.

The medial collateral ligament complex (MCLC) is composed of the anterior oblique, posterior and transverse bands. The anterior band is the primary constraint against valgus stress, contributing 55 to 70% of valgus stability of the elbow. There are two elements of the anterior band: the anterior non-isometric and posterior isometric bundles. There is sequential tightening of these bundles proceeding from anterior to posterior when the elbow is flexed from full extension. The fan-shaped posterior band of the MCLC is a thickening of the capsule that is best defined with the elbow flexed at 90°. The posterior band does not contribute significantly to valgus stability of the elbow except in near terminal flexion. The transverse band

(TB) does not cross the joint, but exists as a thickening of the most caudal portion of the joint capsule. The TB contributes little to elbow stability because it originates from and inserts on the ulna.

The lateral collateral ligament complex (LCLC) consists of four components—the annular ligament, radial collateral ligament, lateral ulnar collateral ligament and accessory lateral collateral ligament—as well as the anconeus muscle (which acts as a dynamic stabiliser). These ligaments provide varus and external rotation stability to the elbow. This ligament complex is isometric throughout the normal range of flexion and extension with little change in the distance between ligament origin and insertion as the ligament complex is near the axis of rotation. There is controversy about the function of the different components of the LCLC.

Muscles

Four muscle groups control elbow dynamic stability: the flexors and extensors of the elbow and the pronators and supinators of the proximal radio-ulnar joint. The flexor and extensor muscles provide some compression across the joint, increasing the inherent stability provided by the congruent articular surfaces.

The brachialis and triceps muscles have broad cross-sectional areas and their insertions are close to the axis of elbow joint rotation, providing a bulk effect to gain further stability of the joint. The overall effect of these muscle contractions on the stability of the elbow is likely to be important, but the magnitude of these effects has yet to be determined.

The muscles that cross the medial side of the elbow joint have an alignment favouring dynamic protection of the MCLC. On the medial side of the elbow the flexor carpi ulnaris (FCU) is the most important dynamic stabiliser, with the flexor digitorum superficialis (FDS) being less important. The FCU and FDS are optimally positioned to provide secondary valgus stability of the medial side of the elbow, especially at between 90 and 120° of elbow flexion (the position of cocking in throwing).

LATERAL ELBOW PAIN

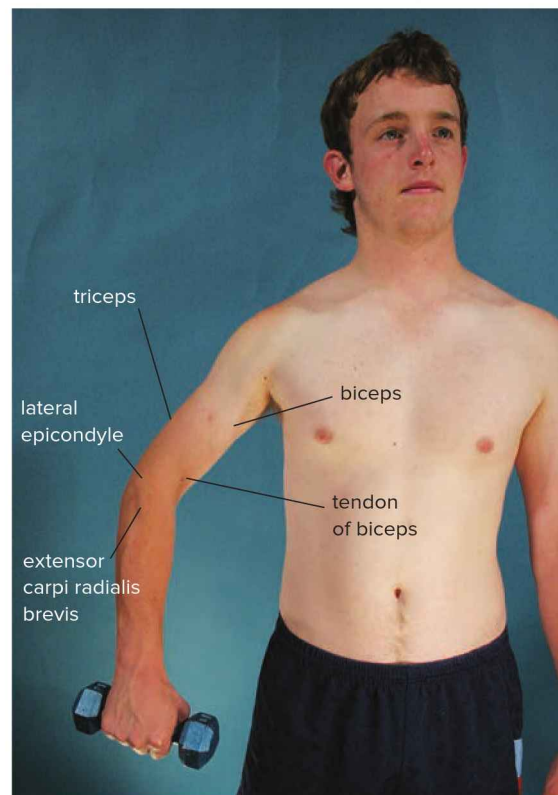
The lateral side is the most common site of elbow pain (Fig. 25.1). Some diagnoses that should be considered are lateral elbow tendinopathy, referred pain from the cervical and upper thoracic spine, synovitis of the radiohumeral joint, radiohumeral bursitis, osteochondritis dissecans of the capitellum and radius, posterolateral elbow instability and combinations of these problems (Table 25.1).

If your patient is between 30 and 60 years of age with local lateral elbow pain with or without some spread into the forearm, but no pain in the neck, arm and beyond the wrist, then lateral elbow tendinopathy is the likely

diagnosis. Although evocative, the term 'tennis elbow' is unsatisfactory as it gives little indication of pathology; in fact, sports medicine clinicians are more likely to see this condition in non-tennis players than in tennis players. The term 'lateral epicondylitis' is not accurate as the primary pathology is not inflammatory as implied by the suffix 'itis'. The term epicondylitis is also problematic, as not all patients will present with degenerative changes implied by the suffix 'osis' (Chapter 4). Lateral elbow tendinopathy or lateral elbow pain are more general terms that do not assume a given pathology or an exact source of symptoms and thus better reflect the clinical situation.

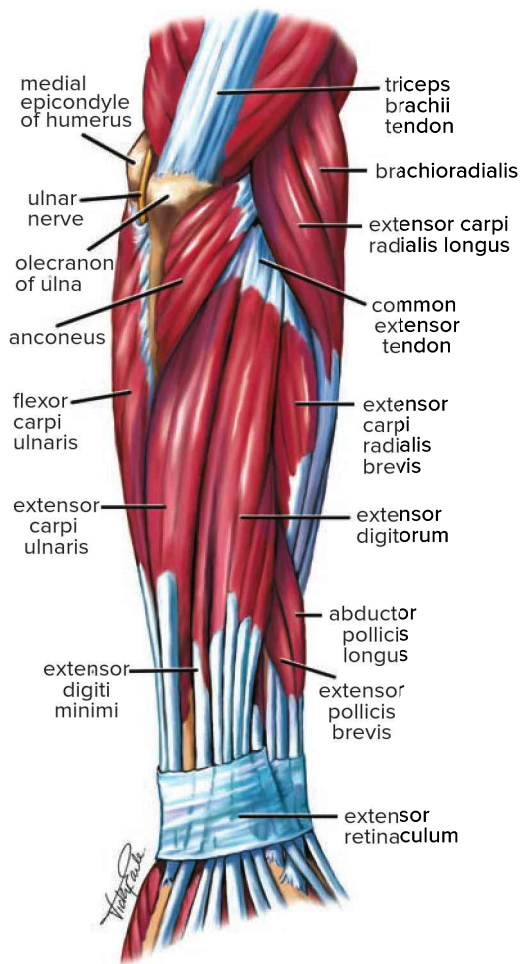
History

Begin by eliciting the characteristics of the patient's lateral elbow pain. The pain of lateral extensor tendinopathy is typically located around the lateral epicondyle and proximal forearm. Occasionally the pain may radiate into the forearm extensor muscle mass, but it never extends into the hand and fingers or proximally into the upper arm. The onset of pain is often insidious but may be acute.



(a)

Figure 25.1 Anatomy of the lateral elbow (a) Surface anatomy of the lateral elbow



(b)

Figure 25.1 (cont.) (b) Anatomy of the lateral elbow from behind

There may have been recent changes in training load, technique, duties or equipment used in sports or work.

The severity of pain ranges from relatively trivial to almost incapacitating, keeping the patient awake at night. It is important to note whether the pain is aggravated

by light everyday activities, such as picking up a cup, or requires repeated heavy activity to become painful.

Pain in the lateral aspect of the forearm may also implicate entrapment of the posterior interosseous nerve or irritation of other nerves. If pain is closely related to the activity level, it is more likely to be of a mechanical origin. If pain is persistent, unpredictable or related to posture, consider a source of pain from other anatomical structures, non-musculoskeletal pathologies or abnormal central nervous system functioning (e.g. complex regional pain or neuropathic pain states). Lateral elbow tendinopathy is typically painful with gripping or wrist extension, whereas referred pain may be provoked by prolonged postures such as in lengthy periods seated at a desk or in a car. Associated sensory symptoms, such as pins and needles, also suggest a neural component. Presence of neck, upper thoracic or shoulder pain should be noted, especially when symptoms extend beyond the lateral elbow and forearm.

Often patients presenting to sports medicine clinicians will have undergone a variety of treatments. It is important to note the response to each of these.

An activity history should also be taken, noting any recent change in the level of activity. In tennis players, note any change in racquet size, grip size or string tension and whether or not any comment has been made regarding the patient's technique. This is also relevant to occupations involving manual tool handling tasks.

Examination

Examination involves:

1. Observation from the front
2. Active movements
 - a. elbow flexion/extension
 - b. supination/pronation
 - c. wrist flexion (forearm pronated) (Fig. 25.2a)
 - d. wrist extension
3. Passive movements
 - a. as above
4. Resisted movements
 - a. wrist extension (Fig. 25.2b)

Table 25.1 Causes of lateral elbow pain

Common	Less common	Not to be missed
Extensor tendinopathy Referred pain • Cervical spine • Upper thoracic spine • Myofascial	Synovitis of the radiohumeral joint Radiohumeral bursitis Posterior interosseous nerve entrapment (radial tunnel syndrome)	Osteochondritis dissecans (in adolescents) • Capitellum • Radius

- b. extension at the third metacarpophalangeal joint (Fig. 25.2c)
 - c. grip test (Fig. 25.2d)
- 5. Palpation
 - a. lateral epicondyle (Fig. 25.2e)
 - b. extensor muscles (Fig. 25.2f)
- 6. Special tests
 - a. neurodynamic tests
 - b. cervical spine examination (Chapter 23)

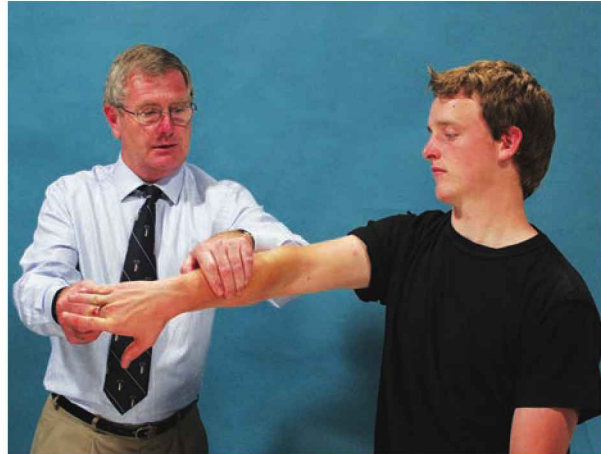
- c. thoracic spine examination (Chapter 28)
- d. periscapular soft tissues (Fig. 25.2g).

Investigations

Investigations are usually not performed in a straightforward case of lateral elbow pain. However, in longstanding cases, plain radiograph (AP and lateral views) of the elbow may show osteochondritis dissecans, degenerative joint changes or evidence of heterotopic calcification.



(a)



(c)



(b)



(d)

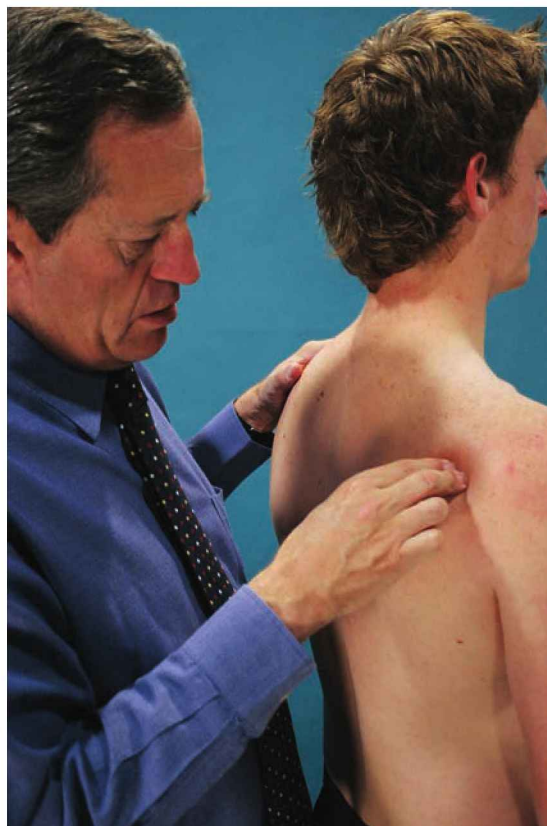
Figure 25.2 Examination of the patient with lateral elbow pain (a) Active movement—wrist flexion with forearm fully pronated (b) Resisted muscle testing—patient resists the clinician's pressure by extension of the wrist (c) Resisted muscle testing—extension at third metacarpophalangeal joint (d) The patient (in sitting or standing position) raises his or her arms up to horizontal, keeping the elbows extended and the palms facing down to the floor, then grips the examiner's two radial fingers as firmly as possible with the unaffected side, followed by the affected side until painful. A positive test is indicated by markedly reduced force to pain onset



(e)



(f)



(g)

Figure 25.2 (cont.) (e) Palpation—lateral epicondyle. Attempt to locate painful site distal to lateral epicondyle (f) Palpation—extensor muscles. Pincer grip is used with passive flexion and extension to provide exact feel of damaged tissue (g) Periscapular soft tissues—palpation of active trigger points and changes in muscle tone and length in the periscapular soft tissues

Ultrasound examination may be helpful.¹ It may demonstrate the degree of tendon or ligament damage² as well as the presence of a bursa; MRI identifies tendinopathy³ but is not indicated routinely. It may provide valuable anatomic detail when the diagnosis is challenging.

Lateral elbow tendinopathy

For this major sports medicine condition, we review the pathology, outline the clinical presentation and then discuss evidence-informed and clinically-reasoned treatment.

Clinically relevant pathology

The extensor carpi radialis brevis (ECRB) tendon is the most common tendon to become problematic, but the extensor digitorum (or extensor carpi ulnaris) may be

implicated if the middle finger extension test is more provocative than the wrist extension test.⁴ In addition to locating which muscle is involved, the tendon may be affected at its origin⁵ or mid-substance. The latter is characterised on light microscopy as an excess of both fibroblasts and blood vessels.⁶⁻⁸ The vessels appear consistent with what Alfredson calls 'neovessels',⁹ pathologists call 'angiogenesis' and Nirschl called 'angiofibroblastic hyperplasia'.⁷ With continued use, the lesions may extend into microscopic partial tears.¹⁰ Conversely, a tear may be the primary abnormality, with degenerative changes being secondary. A summary of the processes leading to extensor tendinopathy is shown in Figure 25.3.

Manual tasks that require wrist stabilisation, such as gripping, or that require wrist extension movements,

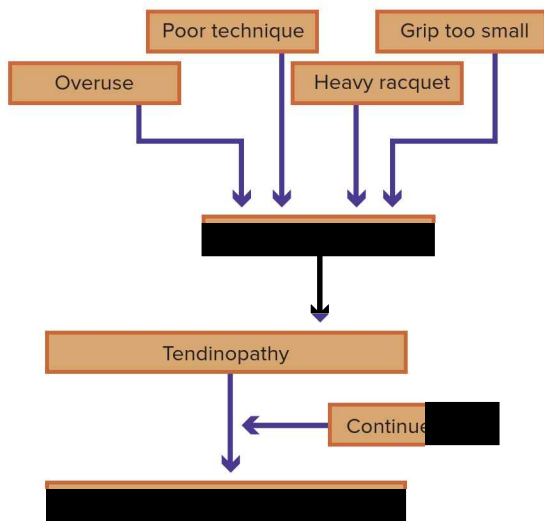


Figure 25.3 Processes leading to the development of lateral elbow tendinopathy

place considerable load on the ECRB tendon. The highest stresses occur in the ECRB tendon when the elbow is extended and the forearm pronated—the most provocative position to reproduce a patient's pain and test grip force.

Pain-free grip force is reduced in lateral elbow tendinopathy and there is altered motor control of the gripping action. In lateral elbow tendinopathy during a low force gripping action, there is a relatively lower activation of the ECRB and a relatively greater activation of the long finger flexors and extensors.¹¹ Associated with this, the wrist is held in a more flexed position.¹² The only muscle that is not weaker in this condition is the extensor digitorum longus.¹³ Taken together, this evidence points to altered control mechanisms of the wrist in lateral elbow tendinopathy.

This abnormal gripping action was also present on the unaffected elbow side in unilateral tendinopathy, suggesting central neural mechanisms at play.¹⁴ Consistent with these findings are impaired reaction time and speed of movement that are bilaterally present in unilateral tendinopathy.^{12, 15} Most intriguingly, tendinosis-like tissue changes can also be triggered bilaterally in response to unilateral overuse as documented in a recent animal model.¹⁴ These discoveries provide a basis for the clinician to be alert to potential central mechanisms and to carefully evaluate, and possibly treat, both sides even when the presentation is unilateral.

Clinical features

Lateral elbow tendinopathy occurs in association with repeated gripping and/or wrist extension actions and

activities. These tasks includes sports, such as tennis, squash and badminton, as well as occupational and leisure pursuits, such as carpentry, bricklaying, sewing and knitting. Keyboard use is also associated with the condition.¹⁶ The peak incidence is between the ages of 40 and 60 years, but the condition may affect any age group, in which case careful attention to differential causes of the symptoms is imperative.

History

There are two distinct clinical presentations of lateral elbow tendinopathy. The most common is an insidious onset of pain, which occurs 24-72 hours after unaccustomed activity involving repeated wrist extension. This occurs typically after a person spends the weekend engaged in manual activity such as laying bricks or using a screwdriver, or after prolonged sessions of sewing or knitting. In the tennis player, it may occur following the use of a new racquet; playing with wet, heavy balls; or overhitting, especially when hitting into the wind. It also occurs when the player is hitting 'late' (getting into position slowly), so that body weight is not transferred correctly and the player relies excessively on the forearm muscles for power (see also discussion of specific tennis biomechanics in Chapter 9).

The other clinical presentation is a sudden onset of lateral elbow pain associated with a single instance of exertion involving the wrist extensors, for example lifting a heavy object, or in tennis players attempting a very forceful single-handed backhand with too much reliance on the forearm and not enough on the trunk and legs. The insidious onset is thought to correspond to microscopic tears within the tendon, whereas the acute onset may correspond to larger macroscopic tendon tears.

Physical examination

On examination, insertional versus mid-substance lesions will be differentiated by the site of maximal tenderness, with the latter being approximately 1-2 cm distal to the lateral epicondyle. Palpation of the entire tendon and the associated muscle(s) will provide valuable information on tissue tightness or hypersensitivity, which is useful in guiding treatment.

Typically, the pain is reproduced by resisted wrist extension, especially with the elbow extended and forearm pronated. Resisted extension of the middle finger is also painful (Fig. 25.2c). It is thought that the ECRB tendon is preferentially stressed in this position as it contracts synergistically to anchor the third metacarpal to allow extension to take place at the digits.

The upper limb neurodynamic or neural provocation test (Chapter 15), especially with radial nerve bias, may

reproduce lateral elbow pain or show restriction of movement compared with the other side. In either situation, this may implicate a neural component to the pain.

Examination of the cervical spine will frequently detect decreased range of movement, especially lateral flexion. The cervical and upper thoracic spine may be stiff to palpation and tender both centrally and over the painful apophyseal joint, usually around the C5–6 level.¹⁷ Active trigger points can occur in the periscapular soft tissues (Fig. 25.2g). In chronic cases, it is not uncommon to find decreased joint play in the elbow joints.

Treatment

No single treatment is totally effective and it is most likely that the patient will be best served through a combination of different treatments selected on the basis of the patient's clinical presentation¹⁸ and preferences, and informed by current best evidence.

The basic principles of treating soft tissue injuries apply, with a specific focus on addressing the grip strength deficit and coordination impairments of the upper limb and specifically at the wrist. These deficits are best addressed through exercise. There must be control of pain (e.g. ice, analgesic medications, relative rest, electrotherapy), encouragement of the healing process, restoration of any flexibility deficit, correction of any predisposing factors and a gradual return to activity.

Treatment of any spinal or neural dysfunctions often speeds up resolution¹⁹ and is highly recommended if there is concomitant neck pain.^{20, 21}

PRACTICE PEARL

Exercise is the key to managing lateral elbow tendinopathy. Other treatments are adjunctive, mainly to resolve pain and facilitate tissue healing.

Exercises for strengthening and coordination

We recommend the following exercises:

- Progressively graduated exercise to improve strength and endurance capacity (Fig. 25.4) as well as normalise flexibility of the forearm muscles (Fig. 25.5).
- Exercises geared at improving coordination (Fig. 25.6) should also be considered.²²

It is unknown how much pain should be experienced during exercise without compromising resolution, but experience indicates that it should be minimised through careful selection of load and type of exercise. For example, in very painful elbows, isometric low load exercise with taping, and/or mobilisation with movements (Fig. 25.7)

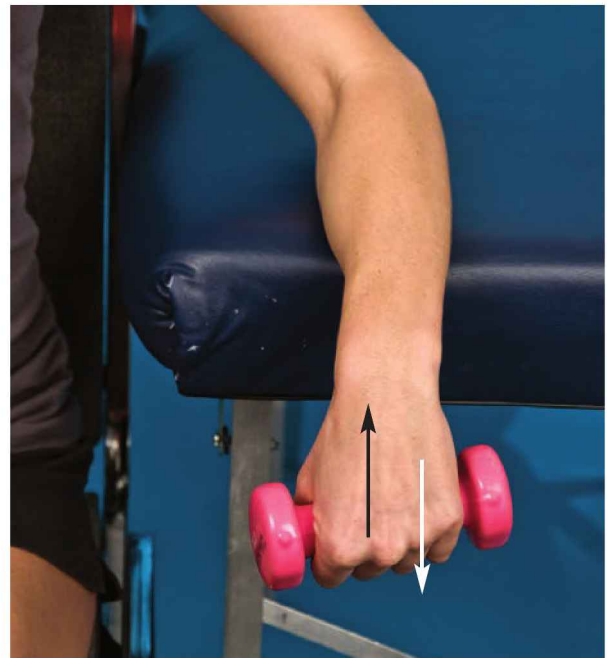


Figure 25.4 Strengthening exercises for wrist extensors.

Exercises can be isometric, concentric (raising the weight, black arrow), eccentric (lowering the weight, white arrow) or functional

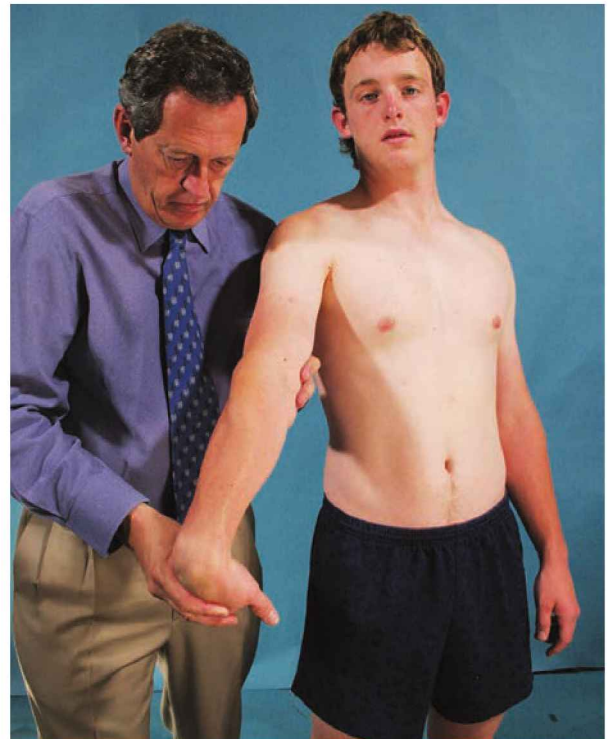


Figure 25.5 Stretching the extensor carpi radialis brevis tendon



Figure 25.6 Coordination exercise focusing on supination/pronation at an early stage in rehabilitation of lateral elbow tendinopathy. The patient oscillates the body blade with the shoulder in a position that approximates the scapular plane



Figure 25.7 Mobilisation with movement

to improve endurance should be favoured initially, with progression in load and complexity (concentric, eccentric) as pain subsides to emphasise strengthening.

Exercise is supported by clinical trials^{23, 24} and in one particular study,²¹ long-term benefits and prevention of recurrence were seen in patients who had previously failed common treatments such as corticosteroid injections and oral medications.

Electrotherapeutic modalities

Electrotherapeutic modalities such as laser, extracorporeal shockwave therapy (ESWT), TENS and ultrasound are often used in practice for these conditions despite mixed evidence on their effectiveness. In some part

related to this mixed evidence is the fundamental issue of what constitutes an appropriate dosage, which can be challenging to the practitioner. For example, laser administered at a wavelength of 904 nm (but not 820 and 1064 nm) offers short-term pain relief and a reduction in disability, applied either in isolation or in combination with exercise.²⁵

A modality where a perplexing situation arises is ESWT, which is effective at some sites of tendinopathy but not for lateral elbow tendinopathy,²⁶ though the issue of dose for this modality has perhaps not been adequately explored. Interestingly, TENS, which is a scientifically developed modality for the reduction of pain, has only recently been considered in clinical trials of tennis elbow.²⁷ Practitioners should be aware that dosage plays an important role in delivering beneficial outcomes;²⁸ high frequencies (e.g. 85–110 Hz) and pulse durations of approximately 200 μ s delivered at a strong but sub-noxious intensity are recommended.

Therapeutic ultrasound (i.e. not diagnostic ultrasound), which tends to be widely used in clinics for many musculoskeletal conditions, has very little evidence to support its use.²⁹ A recent clinical trial showed that while therapeutic ultrasound reduced pain over time, as did the comparator laser treatment, the patients did not rate their condition as being sufficiently improved.³⁰

Manual therapy

Specific manual therapy applied to the joints or soft tissues of the elbow and forearm is beneficial for lateral elbow tendinopathy.

There is evidence of benefit with elbow mobilisation with movement³¹ (Fig. 25.7), in combination with exercise,³² and with the addition of cervical and thoracic spine treatment to elbow treatment.³³

Cervical mobilisation (Fig. 25.8), thoracic mobilisation and neural stretching (Fig. 25.9) are commonly used as adjuncts to other forms of treatment.

Transverse frictions are frequently advocated at the site of the lesion; however, there is little evidence that these are beneficial. There is some evidence that they are less effective than supervised exercise, when frictions are combined with Mill's manipulation, which is a small-amplitude, high-velocity thrust performed at end of elbow extension while the wrist and hand are flexed.^{33–35}

There is little evidence to support the use of digital ischaemic pressures and myofascial massage, though it may be trialled to judge whether the individual patient will benefit from its application. A critical aspect of such soft tissue manual therapy techniques is targeting them to areas exhibiting increased resistance, tightness



Figure 25.8 Cervical mobilisation

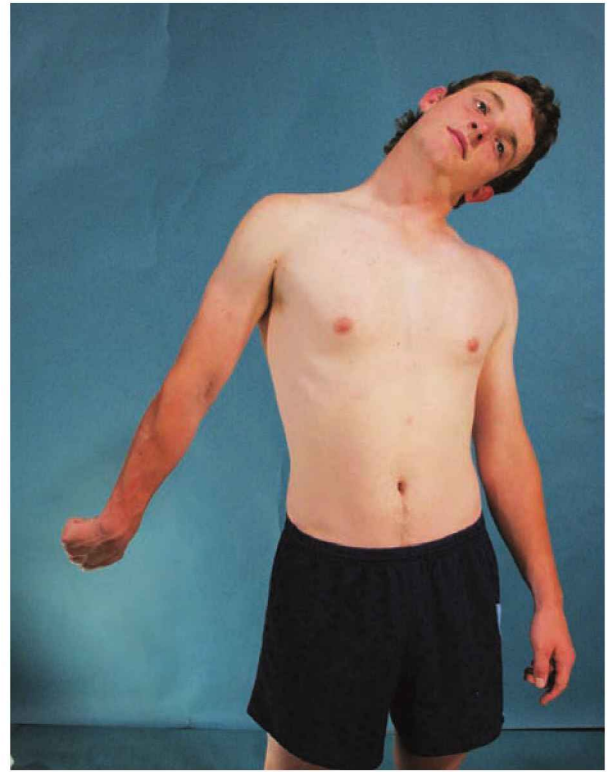


Figure 25.9 Neural tissue mobilisation positioned at end of range

or thickening to digital pressure in the absence of severe pain. In this situation the tissues are usually placed under various conditions of stretch while digital pressure is applied, either sustained (Fig. 25.10a) or longitudinally directed in motion (Fig. 25.10b).

These techniques usually produce pain during application, but this pain should be within the patient's tolerance, not be severe and not outlast the treatment. The presence of severe pain and very little or no resistance, tightness or thickening are contraindications to high force/load soft tissue techniques.

Trigger points

In patients with lateral elbow pain, active trigger points associated with muscle shortening are frequently found in the forearm extensor muscles—brachioradialis, extensor carpi radialis longus, ECRB, extensor digitorum, extensor carpi ulnaris, extensor digiti minimi and anconeus, as well as the periscapular area.³⁶ Digital ischaemic pressure or dry needling of these trigger points may help in the treatment.

Acupuncture

There is some evidence of short-term (2–8 weeks) benefit with the use of acupuncture for lateral elbow pain.³⁷

Bracing and taping

Counterforce bracing (Fig. 25.11) increases forearm extensors' stretching pain tolerance, but it does not appear to influence neuromuscular parameters such as strength or proprioception of these muscles.³⁸ If a brace is to be used, it should demonstrably reduce pain on gripping or on resisted tests of wrist and middle finger extension. In these cases, the brace should be worn during the performance of pain provocative tasks, such as on returning to an aggravating activity like tennis. The brace should be correctly applied approximately 10 cm below the elbow joint. Tensioning the brace up to 50 N has been found to be beneficial.³⁸

A deloading taping technique has been shown to initially improve pain-free gripping³⁹ and to reduce tissue stress within the taped area. Our clinical experience



(a)



(b)

Figure 25.10 Soft tissue techniques (a) Sustained longitudinal pressure to the extensor carpi radialis brevis muscle in the position of maximum elbow extension and wrist flexion (b) Digital ischaemic pressure to deep muscle fibres in the shortened position

indicates that it is very useful in patients who have severe pain and pain at night (Fig. 25.12).

Iontophoresis

There is insufficient evidence from clinical trials to support or refute the use of corticosteroids delivered by iontophoresis.

Corticosteroid injection

PRACTICE PEARL

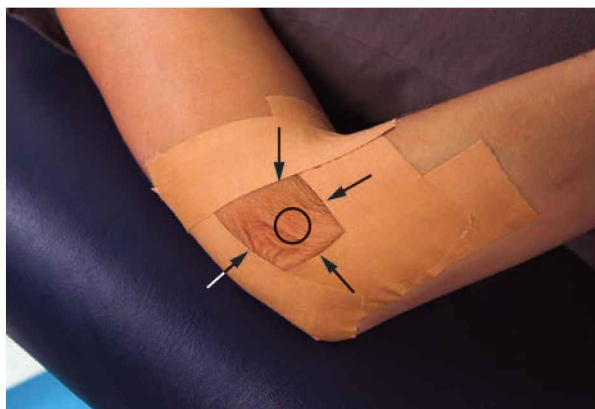
The use of corticosteroid injection in the treatment of this condition requires careful consideration because the best evidence to date indicates that while it is very effective in the short term (i.e. >80% success rate), there are unfavourable consequences later (i.e. >6–12 months), such as delayed recovery and a 62% greater recurrence rate than if the patient were advised to adopt a wait and see approach.⁴⁰



Figure 25.11 Counterforce brace



(a)



(b)

Figure 25.12 The diamond tape technique

(a) The anchor point X marks the site from which the tape is tensioned longitudinally (along the solid-line arrow) and laid onto the skin. The skin should be pulled towards the site of pain (dotted-line arrow)

(b) Note the overlapping ends of the tape and the orange-peel effect on the skin with the diamond tape, resulting in a translation of skin away from the tape towards the site of pain (O) as shown by the arrows

FROM VICENZINO ET AL³⁹

Combining physiotherapy with an injection does not seem to improve this situation.⁴¹ Thus, it is prudent to consider using this injection within a more comprehensive management framework as illustrated (Fig. 25.13).

Prior to any injection, the clinician should fully inform the patient of likely short- and long-term outcomes, as adopting a wait and see policy will result in approximately 80% success rate at 12 months, with much lower risk of recurrences.⁴² When corticosteroid and local anaesthetic agents are injected, the aim should be around the ECRB tendon, directly over the point of maximal tenderness but not into the tendon substance itself.

PRACTICE PEARL

A comprehensive program for the management of lateral elbow pain (Fig. 25.13) would include implementing a graduated progressive exercise program over 8 weeks, pain relief if required, ergonomic advice and activity modification.⁴²

Nitric oxide donor therapy

Initial studies of lateral elbow, Achilles and supraspinatus tendinopathies provided level 2 evidence that nitric oxide donor therapy (glyceryl trinitrate [GTN] patches applied locally, 1.25 mg/day) improved outcomes within

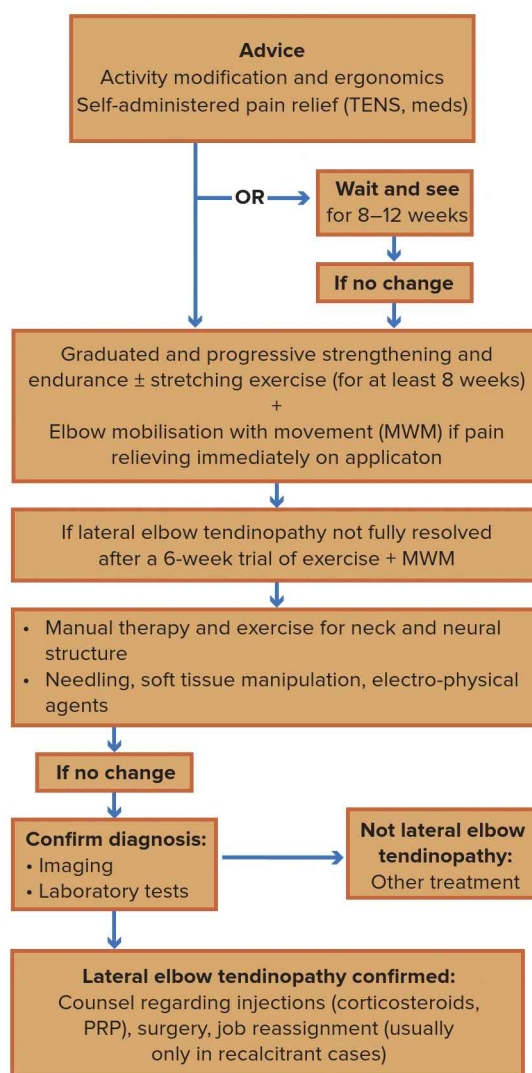
**Figure 25.13** Treatment algorithm



Figure 25.14 Nitric oxide donor therapy—one quarter of a 0.5 mg/24 h glyceryl trinitrate (GTN) patch in place over the most tender site of extensor tendinopathy

3–6 months⁴³ (Fig. 25.14). One mechanism by which this treatment might work is through enhanced collagen synthesis. However, a recent dose-ranging clinical trial for lateral elbow tendinopathy failed to confirm the initial findings, but highlighted that the type of exercise (strengthening as opposed to stretching) is a significant consideration.⁴⁴

Pragmatically, the practitioner and patient need to be aware that 4–5% of patients will develop headaches or skin rash that are severe enough to discontinue treatment with GTN and that this seems to be dose dependent. Thus it is important to carefully meter the dose and monitor response.

Botulinum toxin

Historically used for neuromuscular conditions, botulinum toxin (Botox[®]) injection is a new and unproven treatment of tendinopathy. There is evidence that it produces short-term improvement in pain when compared to placebo,^{40, 45} though it is important to counsel the patient prior to the treatment that there will be a high likelihood (92% at 8 weeks post injection)⁴⁶ of an extensor muscle lag. This treatment-induced extensor weakness is an important consideration for athletes who need good grip strength.

Autologous blood, platelet-rich plasma and autologous cell injections

with ROBERT-JAN DE VOS

Autologous blood and platelet-rich plasma (PRP) are increasingly proposed as treatment for lateral elbow tendinopathy.⁴⁷ The rationale behind these treatments is improvement of tendon healing through delivery of growth factors and cytokines present in blood.^{48, 49} There are various protocols for preparation and injection of these blood products.

A relatively high number of clinical trials have been performed on patients with longstanding lateral elbow tendinopathy. One recent qualitative systematic review, concluded that there is strong evidence against the use of PRP.⁵⁰ Also after pooling of data in a Cochrane review, there was no significant improvement after PRP treatment compared to other treatments for this condition.⁵¹ There is a lot of discussion about the interpretation of these results; nonetheless, the data are clearly not in favour of PRP treatment.⁵² One case series showed pain reduction and functional improvement after an injection with autologous tenocytes for longstanding lateral elbow tendinopathy.^{53, 54} However, due to the study design, this conclusion has limited value.

Consequently, there is currently insufficient evidence to recommend injections with autologous blood, PRP or autologous tenocytes in patients with chronic lateral elbow tendinopathy.

Correct predisposing factors

Probably the most important factor to be avoided is exposure to activity levels for which there is inadequate preparation (e.g. inappropriate training load, poor technique). In tennis players, a major cause is a faulty backhand technique with the elbow leading (Fig. 25.15). Other technique faults that may predispose to the development of extensor tendinopathy include excessive forearm pronation while attempting to hit topspin forehands and excessive wrist flick (flexion) movement while serving. Correction of these faults requires the assistance of a qualified tennis coach. Other factors, such as racquet type, grip size, string tension, court surface and ball weight, may influence the amount of shock imparted to the elbow (Chapter 9). A mid-sized racquet with a large 'sweet spot' and a grip size that feels comfortable should be used. Care should be taken to avoid using racquets with excessively large or, especially, small grips.

Surgery

Surgery might be considered occasionally in cases with a long history (e.g. >18 months) of lateral elbow pain that is recalcitrant to the treatment strategies outlined above. Surgery is varied, but most recent approaches involve ECRB tendon release from the lateral epicondyle with or without some degree of excision of the degenerative tissue within the common extensor tendon. A recent systematic review concluded that there is a lack of quality evidence to recommend one surgical approach over others.⁵⁵

Graduated return to activity

As with all soft tissue injuries, it is important to return gradually to activity following treatment. The tennis



(a)



(b)

Figure 25.15 Backhand technique (a) Incorrect (b) Correct

player should initially practice backhand technique without a ball, then progress slowly from gentle hitting from the service line to eventually hitting full length shots. Depending on the severity of the condition and the

length of the rehabilitation program, this graduated return should take place over a period of at least 3–6 weeks.

Other causes of lateral elbow pain

Other causes of lateral elbow pain may occur in isolation or in conjunction with the previously mentioned conditions. Radiohumeral bursitis is occasionally seen in athletes. This may be distinguished from extensor tendinopathy by the site of tenderness that is over the radiohumeral joint and distal to the lateral epicondyle, maximally over the anterolateral aspect of the head of the radius. The presence of this bursitis may be confirmed on ultrasound examination. Injection with a corticosteroid agent is the most effective form of treatment.

Osteochondritis of the capitulum or radial head may occur in younger athletes (Chapter 44) involved in throwing sports. This is a significant condition as it can cause an enlarged, deformed capitulum that may predispose to the development of osteoarthritis. The treatment of this condition involves avoidance of aggravating activities.

The lateral elbow is a common site of referred pain, especially from the cervical and upper thoracic spine and periscapular soft tissues. Most patients with chronic lateral elbow pain are likely to have some component of their pain emanating from the cervical and thoracic spine (Chapters 23 and 28). Any associated abnormalities of the cervical and thoracic spine should be treated and the patient's signs reassessed immediately after treatment. If there is a noticeable difference, this may indicate a significant component of referred pain.

MEDIAL ELBOW PAIN

Patients who present with medial elbow pain can be considered in two main groups. One group has pain associated with excessive activity of the wrist flexors. This is the medial equivalent of lateral elbow tendinopathy, with a similar pathological process occurring in the tendons of pronator teres and the flexor group. This condition will be referred to as 'flexor/pronator tendinopathy'.

The second group of patients has medial elbow pain related to excessive throwing activities. Throwing produces a valgus elbow stress that is resisted primarily by the anterior oblique portion of the medial collateral ligament (MCL) and secondarily by the stability of the radiocapitellar joint. Repetitive throwing, especially if throwing technique is poor (Chapter 8), leads to stretching of the MCL and a degree of valgus instability. A fixed flexion deformity of the elbow may develop as a result of scarring of the MCL. Subsequently, there may be some secondary impingement of the medial tip of the olecranon onto the olecranon fossa, producing a synovitis or loose

body formation. With valgus stress, the compressive forces may also damage the radiocapitellar joint. Several of these pathological entities may be present in combination.

In children, repetitive valgus stress may result in damage to the medial epicondylar epiphysis with pain and tenderness in this region. This usually responds to a period of rest followed by a gradual return to throwing activity, but may progress to avulsion with continued activity. This condition, commonly known as 'little leaguer's elbow', is considered in Chapter 44. The causes of medial elbow pain are shown in Table 25.2.

Flexor/pronator tendinopathy

This condition is not as common as its lateral equivalent, accounting for 9–20% of all epicondylalgia diagnoses.⁵⁶ It is seen especially in golfers ('golfer's elbow') and in tennis players who use a lot of topspin on their forehand. The primary pathology exists in the tendinous origin of the forearm flexor muscles, particularly in the pronator teres tendon.⁵⁷ Ultrasound is sensitive and specific for detecting clinically defined medial epicondylitis, with focal hypoechoic areas of tendinosis being the most common finding, followed by partial tears—that is, identical to lateral tendinopathy ultrasound findings.

On examination, there is usually localised tenderness just at or below the medial epicondyle with pain on resisted wrist flexion and resisted forearm pronation (Fig. 25.16).

Treatment is along the same lines as treatment of lateral elbow tendinopathy (Fig. 25.13). Particular attention should be paid to the tennis forehand or the golf swing technique. Due to its close proximity to the medial epicondyle, the ulnar nerve may become irritated or trapped in scar tissue. This should be treated with neural mobilisation.

Medial collateral ligament sprain

Sprain of the MCL of the elbow may occur as an acute injury, which is discussed below, or as the result of chronic excessive valgus stress due to throwing. This occurs particularly in baseball pitchers, javelin throwers, and water polo, handball, tennis and cricket players.

Throwing is a highly dynamic activity (see Chapter 8) where body segments move through long arcs of motion at high speed. Large joint forces and torques are generated



Figure 25.16 Medial elbow pain reproduced with resisted wrist flexion and forearm pronation

at the medial side of the elbow. In general, the throwing motion is similar among various sports and the overuse pathology of the medial side of the elbow is similar.

Medial elbow stability between 20 and 120° of elbow flexion is provided primarily by the MCL. The static function of the MCL is to serve as the primary restraint to valgus strains in this functional range of motion. During each throw the elbow is subject to tremendous forces and functional demands. The elbow is designed for planar motion whereas the act of throwing relies on a peak rotatory power to propel the ball.

Biomechanical studies of pitching have shown arm rotation speeds of >7000°/second and arm extension velocity of 2300°/second. The anterior bundle of the MCL is the primary static restraint against valgus stress at 30 to 90° of flexion with an average load to failure of 260 N. Muscle contraction during throwing can reduce the stress on the MCL by compressing the joint and adding some stability. Because of the intensity of activity in these athletes with high numbers of repetitions (in some sports, up to 60 000 throwing motions per year), the abnormal forces over time can lead to subtle microscopic tearing within the soft tissues of the medial side of the elbow with imperfect healing. This chronic attenuation of the MCL with repeated high valgus stress can result in acute rupture of the MCL.

Table 25.2 Causes of medial elbow pain

Common	Less common	Not to be missed
Flexor/pronator tendinopathy Medial collateral ligament sprain • Acute • Chronic	Ulnar nerve compression Avulsion fracture of the medial epicondyle (adolescents) Apophysitis—'Little Leaguer's elbow' (adolescents)	Referred pain

The athlete with an acute MCL injury typically describes the sudden onset of pain with throwing. In approximately 50% of cases the patient will report hearing or feeling a 'pop' and is typically unable to continue throwing. More chronic injuries may not present as obviously.

They will likely be described as a gradual onset of pain localised to the medial elbow that worsens in the late cocking or early acceleration phase of throwing. A decrease in maximum velocity is also typically reported in chronic cases. Local inflammation of the unstable ligamentous complex can lead to other common elbow complaints such as ulnar nerve symptoms secondary to irritation of the nerve within the cubital tunnel (see below), flexor-pronator mass strain or medial epicondylitis.⁵⁸

On examination, there will be localised tenderness over the ligament *just distal to the medial epicondyle* and mild pain/instability on valgus stress (Fig. 25.17a). The *milking manoeuvre* or the moving valgus stress test is a useful test to access partial tear of MCL. The milking manoeuvre is performed by pulling on the patient's thumb in order to apply valgus stress while the patient's shoulder is forward elevated to 90° and elbow flexed beyond 90°.⁵⁹

There will often be associated abnormalities such as a flexion contracture of the forearm muscles, synovitis and loose body formation around the tip of the olecranon, as well as damage to the radio-capitellar joint. However, many throwers demonstrate a flexion contracture without concurrent MCL pathology.⁶⁰ Ultrasound and MRI are the preferred imaging modalities in these athletes.

Treatment in the early stages of the injury involves modification of activity, correction of faulty technique, local electrotherapeutic modalities (just as with lateral elbow tendinopathy, little research exists) and soft tissue therapy to the medial ligament. Medial strapping of the elbow may offer additional protection (Fig. 25.17b). Specific muscle strengthening should be commenced, concentrating on the forearm flexors and pronators (Fig. 25.18).

PRACTICE PEARL

Surgery is indicated for the throwing athlete with elbow pain associated with medial dynamic valgus instability that precludes participation in sport at the pre-injury level.

The first successful UCL reconstruction surgery was performed in 1974 by Dr Frank Jobe on Los Angeles Dodgers pitcher, Tommy John. Prior to this surgery, a UCL tear was considered to be a career-ending injury. Tommy John's return to baseball in 1976 changed the way this injury was viewed and sparked an evolution of surgical techniques that would take place over the next 40 years.



(a)



(b)

Figure 25.17 (a) Assessment of integrity of the medial collateral ligament with humerus fully internally rotated and controlled for pro/supination. Laxity should not exceed 5°
(b) Elbow stability tape

Most MCL reconstruction techniques ('Tommy John surgery') use a free autograft tendon (e.g. palmaris longus) placed in ulnar and humeral bone tunnels. Transmuscular splitting approach to the medial elbow is used to expose the MCL. Time to return to throwing is about 1 year for full competition. Current techniques allow competitive throwing athletes to return to their previous levels of sport in more than 80% of cases.⁶¹

Ulnar nerve entrapment/neuritis

The ulnar nerve pierces the intermuscular septum in the middle of the arm and then passes deep to the medial head of the triceps muscle into a superficial groove (ulnar sulcus) between the olecranon and medial epicondyle. It enters the forearm between the humerus and the ulnar heads of the flexor carpi ulnaris muscle.

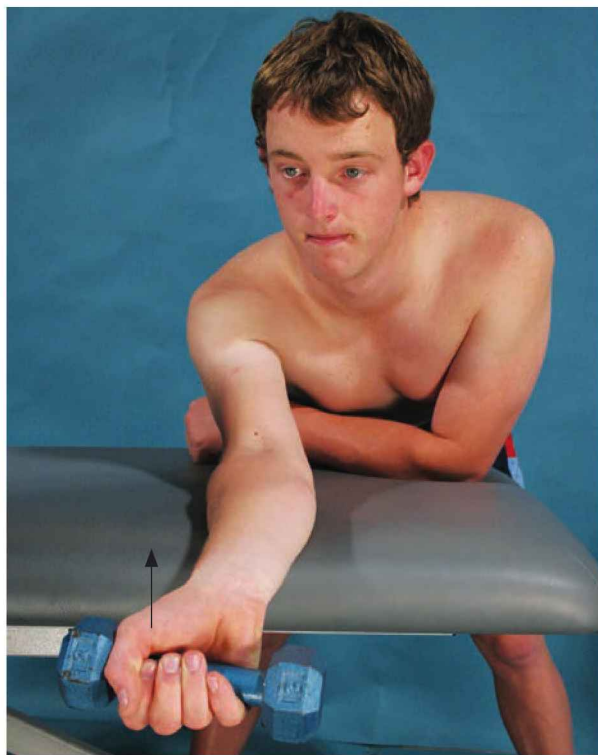


Figure 25.18 Strengthening exercises for the forearm flexors and pronators

Inflammation of the ulnar nerve can occur as a result of a combination of any of four factors.⁶²

1. Traction injuries to the nerve may occur because of the dynamic valgus forces of throwing, especially when combined with valgus instability of the elbow.
2. Progressive compression can occur at the cubital tunnel secondary to inflammation and adhesions from repetitive stresses, or where the nerve passes between the two heads of the flexor carpi ulnaris due to muscle overdevelopment secondary to resistance weight-training exercises.
3. Recurrent subluxation of the nerve due to acquired laxity from repetitive stress or direct trauma. A subluxing ulnar nerve is present in about 8% of the population. Can be an important differential diagnosis—the athlete feels a ‘click’ and might have some pain at the same time.
4. Irregularities within the ulnar groove, such as spurs commonly seen from overuse injuries in older throwers.

The patient presents with posteromedial elbow pain and sensory symptoms such as pins and needles or numbness along the ulnar nerve distribution—the ulnar border of the forearm and the ulnar one and a half fingers. The nerve may be tender behind the medial epicondyle (Fig. 25.19) and tapping over the nerve may



Figure 25.19 Palpation of the ulnar nerve

reproduce symptoms in some cases. Placing the elbow in maximum flexion, the forearm in pronation and the wrist in full extension for one minute may reproduce medial elbow pain and tingling/numbness in the ring and little finger if ulnar neuritis is present. Patients with clinical features of ulnar nerve involvement should undergo nerve conduction studies.⁵⁶ Reports of snapping sensation should lead one to suspect ulnar nerve subluxation, which can be confirmed with dynamic ultrasound examination.⁶³

Treatment of this condition depends on the initiating factor.

1. Traction injuries related to valgus instability from throwing are best served by treating the instability to reduce the ongoing nerve irritation.
2. If adhesions are felt to be present, treatment may include local soft tissue therapy to the nerve in the ulnar groove to mobilise soft tissue that may be compressing or tethering the nerve. Neural mobilisation is often beneficial.
3. Recurrent subluxation of the ulnar nerve should be referred to a neurologist or surgeon experienced in managing this condition in active individuals. Management will depend on the degree of symptoms, electrophysiological evidence of nerve injury and local management expertise (in relation to nerve transposition surgery).
4. Bony irregularities may be amenable to arthroscopic debridement.

POSTERIOR ELBOW PAIN

The main causes of posterior elbow pain are olecranon bursitis, triceps tendinopathy and posterior impingement. Gout should always be considered.

Olecranon bursitis

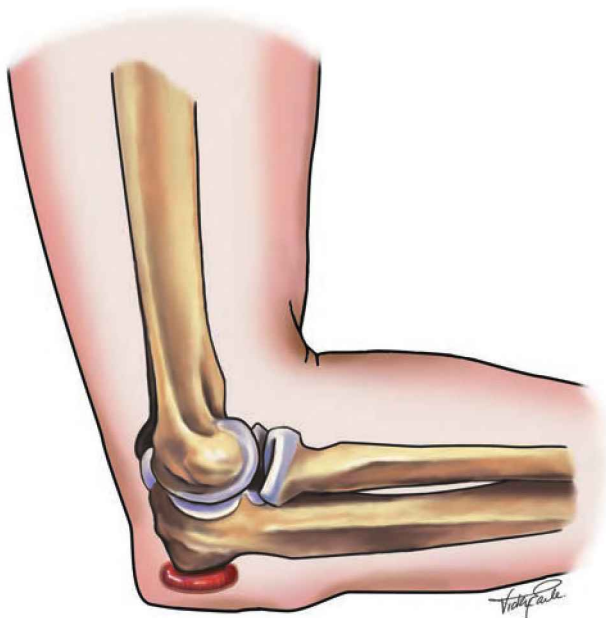
Olecranon bursitis may present after a single episode of trauma or, more commonly, after repeated trauma, such as

falls onto a hard surface affecting the posterior aspect of the elbow. This is commonly seen in football goalkeepers and in basketball players 'taking a charge'. It is also seen in individuals who rest their elbow on a hard surface for long periods of time and is known as 'student's elbow'. The olecranon bursa is a subcutaneous bursa that may become filled with blood and serous fluid (Fig. 25.20).

Treatment in acute traumatic cases consists initially of ice, rest, firm compression and NSAIDs. If this fails, then aspiration of the contents of the bursa and injection with a mixture of corticosteroid and local anaesthetic agents will usually be effective. The needle should be inserted at an oblique angle to reduce the risk of sinus formation.



(a)



(b)

Figure 25.20 Olecranon bursa (a) Palpation site of bursa (b) Olecranon bursitis

Although this is considered a straightforward procedure among experienced clinicians, there is a trend to do this with ultrasound imaging support to increase the accuracy of needle insertion. If recurrent bursitis does not respond to aspiration and injection, surgical excision of the bursa should be considered.

Occasionally, olecranon bursitis can become infected. This is a serious complication that requires immediate drainage, immobilisation and antibiotic therapy. Osteomyelitis and septic arthritis can follow. Excision of the bursa is occasionally required.

Triceps tendinopathy

Tendinopathy at the insertion of the triceps onto the olecranon is occasionally seen. Standard conservative measures for treatment of tendinopathy should be used. Soft tissue therapy including self-massage with a Styrofoam™ roll and dry needling to reduce excessive tightness of the triceps musculotendinous complex may be helpful.

Posterior impingement

Posterior impingement is probably the most common cause of posterior elbow pain. It occurs in three situations. In the younger athlete there is the 'hyperextension valgus overload syndrome'. Repetitive hyperextension valgus stress to the elbow results in impingement of the posterior medial corner of the olecranon tip on the olecranon fossa. Over time this causes osteophyte formation, exacerbating the impingement and leading to a fixed flexion deformity. Valgus instability can be a primary driver of posterior impingement as the olecranon will then no longer be fitting into the olecranon fossa, but will impact more medially.

In the older patient the most common cause is early osteoarthritis, which often predominantly affects the radiocapitellar joint. Generalised osteophytes form through the elbow. Impingement of these osteophytes posteriorly results in posterior pain. The main clinical feature in athletes with posterior impingement is a fixed flexion deformity of some degree and posterior pain with forced extension (Fig. 25.21).

Treatment may include strategies to minimise hyperextension forces such as taping or bracing, along with a strength and flexibility program and graduated return to sport or activity. If lost range is the only problem, surgery is rarely indicated.

If conservative measures fail, then arthroscopic removal of the impinging posterior bone and anterior capsule release is very effective in relieving symptoms and improving extension especially in elderly patients. Acute painful locking however is an indication for surgery. The patient must be counselled that sometimes the loose bodies can be hard to find in the elbow, and the surgeon may miss one or more.



Figure 25.21 Posterior impingement. The elbow is forced into end-range extension. If posterior pain is produced then posterior impingement is present

ACUTE ELBOW INJURIES

Acute elbow injuries include fractures, dislocations and ligament or tendon ruptures. Elbow dislocation associated with fracture can lead to long-term instability and must be recognised early.

Fractures

As the complication rate for elbow fractures is higher than with fractures near other joints, it is essential that fractures in this region be recognised and treated early and aggressively. Those who are fully able to extend the elbow are unlikely to have a fracture, though they should be followed up in 7–10 days if symptoms have not resolved. A clinical protocol has been proposed to detect the presence of a fracture.⁶⁴ Plain radiography is the first line of investigation.

PRACTICE PEARL

Patients who cannot fully extend their elbow after injury should be referred for plain radiography because there is a 50% chance of fracture.

Unstable fractures, usually those associated with displacement, should be referred early for orthopaedic management. When the articular or cortical surface has less than 2 mm of vertical or horizontal displacement, the fracture can be regarded as stable and treated conservatively.⁶⁵

The most common complication of elbow fractures is stiffness, particularly loss of terminal extension. Prompt diagnosis and treatment that includes an early rehabilitation program can help avoid loss of extension. Thus, treatment of

elbow fractures must be aggressive. Surgically stabilising a displaced adult elbow fracture allows early commencement of a post-operative range of motion program.

A stable fracture that involves no significant comminution, displacement or angulation may be treated conservatively. In adults, immobilising the arm for a few days, even up to a week, is generally well tolerated. Then the arm should be placed in a removable splint and early motion commenced. The fracture should then be protected for 6–8 weeks, with early and frequent radiographic checks to ensure the reduction stays anatomical.

Heterotopic ossification is the other main complication of elbow fractures, particularly in high energy injuries. This usually appears within the first month after surgery and plateaus after 4–6 months. Traumatized elbows that are forcefully or passively manipulated may also be at greater risk of this complication.⁶⁶ Therefore, gentle, active assisted range of motion and pain-free stretching exercises are preferred. Mobilisation with movements, applied correctly in a pain-free manner, may be helpful in restoring motion.³¹ Heterotopic bone formation has also been associated with elbow fracture-dislocation treated surgically 1–5 days after injury or treated with multiple surgical procedures. Thus, surgery should be performed in the first 24 hours after injury or after 5–7 days.

Supracondylar fractures

Supracondylar fractures are more common around the age of 12 than in adults. They often occur from a fall on an outstretched arm, either from a height or a bicycle. As they are rotationally unstable and have a high rate of neurovascular complications, these fractures should be regarded as an orthopaedic emergency.

For fractures that are unstable, displaced or cannot be reduced without jeopardising the blood supply, the treatment of choice is closed reduction in the operating room under general anaesthesia. Percutaneous pins placed across the fracture maintain the reduction and prevent late slippage. The arm is initially placed in a splint and then several days later in a cast. The pins are removed after 4–6 weeks. Stiffness is typically not a problem in children recovering from fractures.

Olecranon fractures

Olecranon fractures occur from a fall onto an outstretched hand or from direct trauma to the elbow. If the fracture is non-displaced and stable, the patient should be able to extend the arm against gravity. Treatment consists of immobilising the arm for 2–3 weeks in a posterior splint and then in a removable splint, and commencing a range of motion program. If the patient is unable to extend the elbow against gravity or if radiographs show significant displacement, open reduction with internal fixation by

tension-band wiring is preferred.⁶⁷ Early motion is started within one week of surgery.

Radial head fracture

The most common fracture around the elbow in athletes is the radial head fracture, almost always resulting from a fall onto an outstretched hand. The radial head can be considered a multifunctional stabiliser of the elbow. In the frontal plane its contribution to valgus stability may be as high as 30% even with the intact MCL. In disruption of the MCL, the radial head becomes a critical secondary stabiliser of the elbow with up to 70% of resistance to valgus stress. In the sagittal plane, the radial head contributes to preventing posterior elbow dislocation in association with the coronoid process, lateral and medial ligament complex. In the longitudinal plane, the radial head prevents proximal migration of the radial shaft along with the interosseous membrane.

The main mechanism of injury in contact sports is a fall onto the outstretched hand—the radial head may fail under compressive axial or valgus loads. If there is a rotatory component it may cause a marginal shearing fracture. High energy injuries are more likely to provoke an associated fracture (capitellum, coronoid or even olecranon). MCL ligament injuries are very frequently associated.

AP, lateral and oblique elbow radiographs usually provide sufficient information for diagnosis and deciding on treatment. CT scan of the elbow in some cases can be very useful if surgery is needed, or in doubtful cases after plain radiography (for more complex cases).

Radial head fractures have been classified by Mason.⁶⁸

- Type I—undisplaced
- Type II—displaced wedge fragment
- Type III—comminuted fracture
- Type IV—fracture associated with elbow dislocation.

Most radial head fractures are minimally displaced or non-displaced (type I) and are very difficult to see on radiographs. Sometimes the only clue is the fat pad sign, which appears as a triangular radiolucency just in front of the elbow joint. Early aspiration, splinting with an easily removable device and early commencement of a range of motion program will yield excellent results. Complete healing can be expected within 6–8 weeks.

Open reduction and internal fixation of displaced type II radial head fractures should only be attempted when anatomic reduction, restoration of articular congruity and initiation of early motion can be achieved. If these goals are not obtainable, open reduction and internal fixation may lead to early fixation failure, non-union, and loss of elbow and forearm motion and stability.

Comminuted radial head fractures are frequently associated with elbow dislocation and other destabilising

injuries.⁶⁹ The scarce literature on the subject is of little help in defining optimal treatment.⁷⁰ Radial head excision in young or adult athletes with comminuted fracture of the radial head, associated with medial, lateral or interosseous ligament injury, has a poor clinical outcome. There is currently more emphasis on preservation or replacement of the radial head, attempting to avoid radial head excision if at all possible.^{70, 71}

Coronoid fractures

The coronoid process provides an anterior bony buttress to resist the posteriorly directed forces that occur from the flexor/extensor muscles. Regan and Morrey in their study⁷² showed a correlation between coronoid fragment size and a tendency for elbow dislocation.

Coronoid fracture can be isolated, but most commonly occurs in association with radial head fracture and elbow dislocation ('terrible triad'). The injury pattern is determined by the position at the time of injury and the loading pattern across the elbow. A common mistake is to misidentify a coronoid fragment as a radial head fragment. Coronoid fractures have been classified on the base of anatomic location of the primary fracture:

- tip (anteromedial facet)
- body.

Plain AP, lateral and oblique lateral radiograph views can be helpful and CT scan is highly useful.

PRACTICE PEARL

Dislocated elbow with radial head fracture usually has an associated coronoid fracture. The coronoid fragment is always bigger than it appears on the radiograph because of the articular cartilage cup on the fragment.

Caution is required when treating isolated coronoid fractures as they appear benign, but usually are fracture subluxations with ligament avulsion. Many of these fractures require some form of surgical treatment by specialised elbow surgeons.

Posterior dislocation

The most serious acute elbow injury is a posterior dislocation. This can occur either in contact sports or when falling from a height such as while pole vaulting. There is often an associated fracture of the coronoid process or radial head. The usual mechanism is a posterolateral rotatory force resulting from a fall on an outstretched hand with the shoulder abducted, axial compression, forearm in supination and then forced flexion of the elbow.⁷³

There are three grades of elbow dislocation:⁷³

1. subluxation of the elbow in the posterolateral direction
2. incomplete elbow dislocation with the coronoid perched on the trochlea
3. complete posterior dislocation of the elbow, so that the coronoid rests behind the trochlear
 - a. where the anterior band of the MCL is intact and the elbow is stable to valgus stress following reduction
 - b. where the MCL is completely disrupted so the elbow is unstable in all directions after reduction.

This pattern of dislocation is associated with progressive soft tissue disruption that commences laterally and progresses around anteriorly, medially and posteriorly, termed the Horii circle described by O'Driscoll.⁷⁴

The major complication of elbow dislocation is impairment of the vascular supply to the forearm. Assessment of pulses distal to the dislocation is essential. If pulses are absent, reduction of the dislocation is required urgently. Reduction is usually relatively easy. With the elbow held at 45°, the clinician stabilises the humerus by gripping the anterior aspect of the distal humerus and traction is placed longitudinally along the forearm with the other hand (Fig. 25.22). The elbow usually reduces with a pronounced clunk. If vascular impairment persists after reduction, urgent surgical intervention is required.

Following reduction, the stability of the collateral ligaments should be assessed (Fig. 25.17a). A post-reduction radiograph should also be performed. Small fractures of the coronoid process or undisplaced fractures of the radial head only require conservative treatment with support in a sling for 2-3 weeks. Large coronoid fractures, however, may result in chronic instability and should be reduced and fixed surgically. Large fractures of the radial head may be difficult to manage, but in most cases can be internally fixed. Occasionally, a large fracture of the capitellum may



Figure 25.22 Technique for reduction of posterior dislocation of the elbow

occur. This also requires internal fixation. Sometimes a piece of bone becomes trapped in the joint after reduction. This needs to be excluded with good quality post-reduction radiographs.

Acute injuries at the elbow after subluxation or dislocation are best treated non-surgically, with brace or splint keeping the forearm in pronation to allow the lateral ligament complex to heal. Long-term loss of terminal extension is frequently a problem following elbow dislocation. Immediate active mobilisation under supervision has been shown to result in less restriction of elbow extension with no apparent increase in instability.⁷⁵ Professional athletes with a simple dislocation with no associated fracture or instability are able to return to sports relatively quickly after an appropriate rehabilitation program. Verrall described three cases of stable dislocations in professional footballers who returned to sport after 7, 13 and 21 days with no further complications.⁷⁶ Joint mobilisation (Fig. 25.23) may be required as part of the treatment.³¹ The surrounding muscles should also be strengthened. Elbow stability taping should be applied on return to the sport (Fig. 25.17b).

Heterotopic ossification occasionally occurs following elbow dislocation. The use of NSAIDs for a period of three months following the injury may reduce the incidence of this complication.

Elbow dislocations in directions other than posterior occur occasionally. These are often associated with severe ligamentous disruption and patients should be referred to an orthopaedic surgeon immediately.

Instability post-dislocation

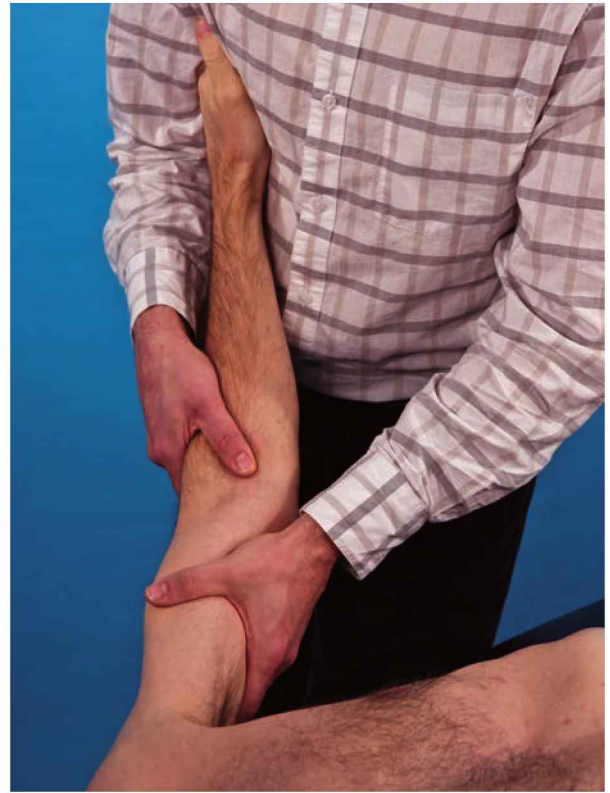
When the competency of the soft tissue restraints after elbow dislocation or injury is inadequate, it can lead to recurrent elbow instability. The condition is more common in some sports than has previously been recognised. On the lateral side, instability is posterolateral and rotatory. On the medial side it is valgus.

Athletes with posterolateral rotatory instability (PLRI) usually present with the symptoms after an elbow subluxation or dislocation. Less commonly there is a history of previous lateral elbow surgery (iatrogenic) for tennis elbow. Recurrent lateral instability is always rotatory.⁷⁷ PLRI is defined as a posterior radial head subluxation relative to the humerus while maintaining normal relationship at the proximal RU joint. O'Driscoll et al. identified the ulnar part of the lateral collateral ligament as the key lateral soft tissue restraint preventing posterolateral rotatory instability;⁷⁷ however, the radial collateral ligament also contributes to both varus and posterolateral stability.

The patient with PLRI presents with recurrent painful clicking, snapping, clunking or locking of the elbow occurring in extension of the arm with the forearm in



(a)



(b)

Figure 25.23 Examples of mobilisation with movement techniques that use a glide out of the plane of extension to improve extension (a) Sustained lateral glide applied through a belt while assisted active extension is performed (b) Sustained internal rotation

supination. Typically these symptoms are more prominent with loading of the joint in the supinated and slightly flexed position. Several provocative manoeuvres have been described, including the posterolateral rotatory apprehension test (or lateral pivot shift test)⁷⁷ and the chair and push-up tests.⁷⁸

Chronic PLRI does not respond well to conservative treatment and usually requires ligament reconstruction with a free ligament autograft (ipsilateral palmaris longus, gracilis, semitendinosus, triceps). Published studies on the clinical outcome after surgical treatment of chronic PLRI demonstrate mostly good and excellent outcomes.^{79, 80}

Acute rupture of the medial collateral ligament

Acute rupture of the MCL may occur in a previously damaged ligament or in a normal ligament subjected to extreme valgus stress, for example elbow dislocation. In a study of 14 MCL injuries in American National Football League (NFL) gridiron players,⁸¹ the two most common mechanisms of injury were blocking at the line

of scrimmage (50%) and the application of a valgus force with the hand planted on the playing surface (29%). The degree of laxity should be assessed by applying valgus stress to the elbow at 30° of flexion (Fig. 25.17a). If complete disruption is present with associated instability, surgical repair of the ligament is required. Incomplete tears should be treated with protection in a brace and muscle strengthening exercises for a period of 3–6 weeks, followed by graduated return to sports.

Tendon ruptures

Acute avulsion of the biceps or triceps tendons from their insertions is a rare condition. Rupture of the distal biceps tendon insertion occurs predominantly in young or middle-aged men engaged in strength activities (e.g. weightlifting). Partial ruptures are more painful than complete ruptures, due to mechanical irritation of the remaining intact tendon. Early surgical repair of complete ruptures would be expected to lead to better outcomes, as complete rupture of tendons leads to degenerative processes due to the loss of mechanical load. Rupture of the triceps tendon occurs

most commonly with excessive deceleration force, such as occurs during a fall or by a direct blow to the posterior aspect of the elbow. Partial and complete triceps ruptures are seen in NFL linemen.³⁴ Partial tears tend to heal well without surgery. Acute complete ruptures at the insertion of either of these tendons should be treated surgically.

Hyperextension injuries

Hyperextension injuries caused by direct force to the forearm pushing the elbow into hyperextension may be acute, in which case the athlete is usually unable to continue activity, or repetitive, when the athlete feels sudden pain but is able to continue. These injuries are probably more common than previously thought.

Kenter et al. reported that hyperextension was the most common (56%) cause of acute elbow sprain in NFL players.⁸¹ Hyperextension injuries are common (75%) in goalies in handball.^{82, 83} Hyperextension of the elbow is caused by the ball hitting the fully extended distal part of the forearm when blocking a shot. This injury is frequently bilateral. Most handball goalkeepers could usually remember their first elbow accident. The goalies routinely minimise their symptoms in an attempt to continue to play, despite the fact that the injury affects their performance. Their first symptoms (pain) usually began acutely but developed into chronic elbow pain, with intermittent pain as the chief complaint. There are reports of similar mechanisms of injuries in other sports such as water polo players, soccer goalkeepers, volleyball players, wrestlers and judo practitioners.

The force of the blow of the ball to the forearm produces a valgus load on the elbow, applying stress to the anterior band of the MCLC. Specific injuries include (i) lesions of the anterior capsule including rupture, (ii) L-shaped rupture of the flexor/pronator origin with elongation of the anterior part of the MCL, (iii) occasional incomplete rupture of the LCL and, rarely, (iv) detachment of small fragments of cartilage near the posterior edge of the olecranon. Imaging studies (ultrasound/MRI) can confirm the diagnosis.

FOREARM PAIN

Fracture of the radius and ulna

The bones of the forearm are commonly injured by a fall on the back or front of the outstretched hand. It is usual for both bones to break, although a single bone may be fractured in cases of direct violence or in fractures of the distal third where there is no shortening.

A displaced fracture is usually clinically obvious. Plain radiographs should be taken for post-reduction comparison and for exclusion of a concurrent dislocation. Two types of dislocation occur—the Monteggia injury (fractured ulna with dislocated head of the radius at the elbow joint) and

the Galeazzi injury (fractured radius with dislocated head of the ulna at the wrist joint).

In the child, angulation of less than 10° is acceptable. Other fractures should be reduced under local or general anaesthesia depending on the age of the child. The usual position for immobilisation is in pronation, although in proximal radial fractures and in Smith's fracture at the wrist, the forearm should be held in supination. The plaster should extend above the elbow and leave the metacarpophalangeal joints free. Depending on the age of the child, immobilisation should last 4–6 weeks. The position should be checked by radiography every 1–2 weeks depending on stability.

In the adult, perfect reduction of radial and ulnar fractures is necessary to ensure future sporting function. Most of these fractures are significantly displaced and require anatomical internal fixation with a plate and screws. Depending on the accuracy of reduction, either a cast or a crepe bandage support is required post-operatively for 8–10 weeks. Isolated fractures of the ulna are treated conservatively by an above-elbow cast in mid-pronation for 8 weeks. Monteggia and Galeazzi injuries are usually displaced and should be referred to an orthopaedic surgeon for reduction.

Stress fractures

Stress fractures of the forearm bones occur occasionally in athletes involved in upper limb sports, such as baseball, tennis or swimming. Treatment involves rest and correction of the possible predisposing factors, such as faulty technique.

Entrapment of the posterior interosseous nerve (radial tunnel syndrome)

The radial nerve divides into the superficial radial and the posterior interosseous nerve at the level of the radiocapitellar joint (Fig. 25.24). The posterior interosseous nerve (PIN) passes distal to the origin of the ECRB and enters the arcade of Frohse. Prior to entering the arcade of Frohse, it gives off branches to the ECRB and supinator muscles. The arcade is a semicircular fibrous arch at the proximal head of the supinator muscle, which begins at the tip of the lateral epicondyle and extends downwards, attaching to the medial aspect of the lateral epicondyle. The PIN then emerges from the supinator muscle distally, where it divides into terminal branches that innervate the medial extensors. Compression of the PIN may occur at one of four sites:⁸⁴

- fibrous bands anterior to the radial head
- recurrent radial vessels
- arcade of Frohse
- tendinous margin of the extensor carpi radialis brevis muscle.

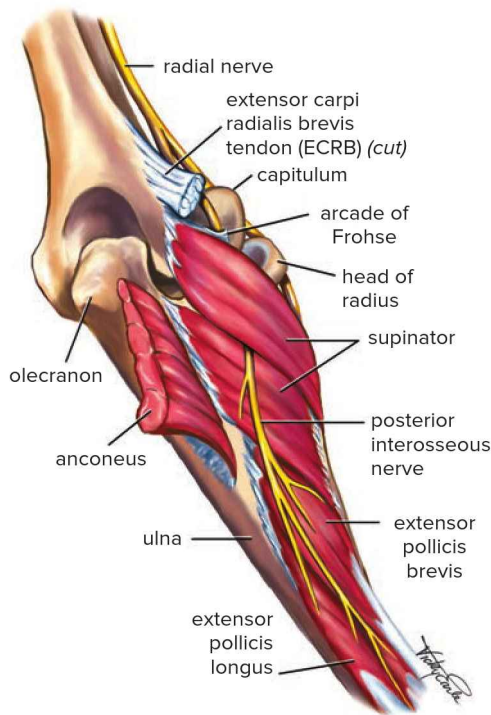


Figure 25.24 Anatomy of the posterior interosseous nerve

It may be difficult to differentiate between lateral elbow tendinopathy and the early stages of PIN entrapment. PIN entrapment is seen in patients who repetitively pronate and supinate the forearm, whereas lateral elbow tendinopathy is more frequently associated with repetitive wrist extension. Symptoms of PIN entrapment include paraesthesia in the hand and lateral forearm, pain over the forearm extensor mass, aching wrist and middle or upper third humeral pain. It is these symptoms that critically differentiate this condition from lateral elbow tendinopathy.

Maximal tenderness is over the supinator muscle, four finger breadths below the lateral epicondyle (distal to the area of maximal tenderness in extensor tendinopathy). Reproduction of symptoms by manual palpation of these local structures and the relief of such palpation-induced symptoms by injection of local anaesthetic should be considered as part of the physical examination.⁸⁵ Nerve entrapment also causes marked pain on resisted supination of the forearm with the elbow flexed to 90° and the forearm fully pronated. Neurodynamic tests (with radial nerve bias) may reproduce the patient's symptoms, and nerve conduction studies may be performed to confirm the diagnosis.

Treatment consists of soft tissue therapy over the supinator muscle at the site of entrapment and neural tissue mobilisation, along with exercises targeting strength and endurance deficits in the forearm muscles. If this is unsuccessful, decompression surgery may be required.

Pronator teres syndrome (median nerve entrapment)

The median nerve can become entrapped as it passes between the two heads of the pronator teres muscle. Median nerve entrapment presents as diffuse anterior elbow pain that radiates distally into the forearm with paraesthesia in the median nerve distribution distally and weakness of the thenar muscles. The most common cause in athletes is forceful pronation, for example throwing or racquet sports.

A similar clinical picture can be produced by cervical radiculopathy at C6–7 level, compression of the median nerve underneath the lacertus fibrosis or by the ligament of Struthers. The diagnosis can be confirmed on nerve conduction studies.

Activity modification is the first line of treatment. Surgical resection of the humeral head of pronator teres is occasionally required.

Forearm compartment pressure syndrome

Forearm compartment pressure syndromes have been described in kayakers, canoeists, motorcyclists (popularly termed 'arm pump' in motorcross) and weight-training athletes. The flexor compartment is most usually affected. Symptoms include activity-related pain that is relieved by rest. Diagnosis requires compartment pressure testing. Treatment consists of local soft tissue therapy. Surgical fasciotomy may be required.

UPPER ARM PAIN

An aching pain in the upper arm is a common complaint, especially among manual workers (e.g. bricklayers, carpenters) and athletes. The most common cause is myofascial pain, but stress fracture of the humerus needs to be considered.

Myofascial pain

A dull non-specific pain in the upper arm is most likely to be myofascial in nature. The most common source of the upper arm pain is trigger points in the infraspinatus muscle (Fig. 25.25). Firm palpation of these trigger points will often reproduce the patient's pain. The cervical spine and glenohumeral joint need to be assessed for their possible involvement and treatment directed accordingly.

Treatment consists of digital ischaemic pressure or dry needling to the trigger points. Attention should also be paid to the lower cervical and upper to mid-thoracic spine. Increased muscle tone and trigger points may be found in the paraspinal muscles, and hypomobility of the intervertebral segments may be present. These abnormalities must also be treated with soft tissue techniques.

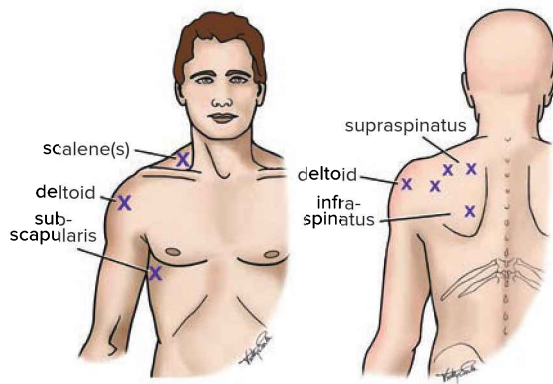


Figure 25.25 Myofascial trigger points around the shoulder region that refer pain to the upper arm

Stress reaction of the humerus

Stress reactions and fractures of the humerus have been described in tennis players, javelin throwers, baseball pitchers, bodybuilders and weightlifters. In a group of symptomatic elite tennis players, MRI of the humerus demonstrated bone marrow oedema and/or periostitis, and the extent of imaging changes was related to the severity and duration of symptoms.⁸⁶ Most of the fractures occurred in adolescents and were associated with a recent increase in activity. In a number of cases the diagnosis was made retrospectively when an acute fracture occurred and the patient admitted symptoms leading up to the acute episode.

Recommended treatment follows the general principles of management of simple stress fractures: avoidance of the aggravating activity until symptom-free with no local tenderness, then gradual resumption of the activity.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e